

# TF-IDF

Term Frequency, Inverse Document Frequency. Intuition, then advantages and disadvantages.

# Every Present Word Gets the Same Value

DATASET · CLEANED

text

words

S1 phone fast2

S2 camera fast sharp3

S3 phone camera fast3

VOCABULARY · IN HOW MANY SENTENCES

fast3 of 3

phone2 of 3

camera2 of 3

sharp1 of 3

| BOW VECTORS |      |       |        |       |
|-------------|------|-------|--------|-------|
|             | fast | phone | camera | sharp |
| S1          | 1    | 1     | 0      | 0     |
| S2          | 1    | 0     | 1      | 1     |
| S3          | 1    | 1     | 1      | 0     |

"fast" is in **every** sentence, yet it gets the same 1 as rarer words. TF-IDF gives each word a **weight**.

TWO PARTS

# Term Frequency × Inverse Document Frequency

**DATASET · CLEANED**

|    | text              | words |
|----|-------------------|-------|
| S1 | phone fast        | 2     |
| S2 | camera fast sharp | 3     |
| S3 | phone camera fast | 3     |

**VOCABULARY · IN HOW MANY SENTENCES**

fast  
3 of 3

phone  
2 of 3

camera  
2 of 3

sharp  
1 of 3

$$\text{TF} = \frac{\text{No. of times the word repeats in the sentence}}{\text{No. of words in the sentence}}$$

$$\text{IDF} = \ln \frac{\text{No. of sentences}}{\text{No. of sentences containing the word}}$$

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

STEP 1

# Term Frequency (TF)

DATASET · CLEANED

|    | text              | words |
|----|-------------------|-------|
| S1 | phone fast        | 2     |
| S2 | camera fast sharp | 3     |
| S3 | phone camera fast | 3     |

VOCABULARY · IN HOW MANY SENTENCES

fast  
3 of 3

phone  
2 of 3

camera  
2 of 3

sharp  
1 of 3

TF = times word repeats in sentence ÷ words in sentence

|    | fast | phone | camera | sharp |
|----|------|-------|--------|-------|
| S1 | 1/2  | 1/2   | 0      | 0     |
| S2 | 1/3  | 0     | 1/3    | 1/3   |
| S3 | 1/3  | 1/3   | 1/3    | 0     |

S1 has 2 words, so present words get 1/2. S2 and S3 have 3 words, so 1/3.

# Inverse Document Frequency (IDF)

DATASET · CLEANED

|    | text              | words |
|----|-------------------|-------|
| S1 | phone fast        | 2     |
| S2 | camera fast sharp | 3     |
| S3 | phone camera fast | 3     |

VOCABULARY · IN HOW MANY SENTENCES

fast  
3 of 3

phone  
2 of 3

camera  
2 of 3

sharp  
1 of 3

IDF = ln( sentences ÷ sentences containing the word )

| WORD   | IDF                |
|--------|--------------------|
| fast   | $\ln(3/3) = 0$     |
| phone  | $\ln(3/2) = 0.405$ |
| camera | $\ln(3/2) = 0.405$ |
| sharp  | $\ln(3/1) = 1.099$ |

In every sentence → 0. Rarer word → bigger IDF.

STEP 3

Final TF-IDF = TF × IDF

IDF

fast

0

phone

0.405

camera

0.405

sharp

1.099

DATASET

S1 phone fast

S2 camera fast sharp

S3 phone camera fast

|    | fast | phone                                             | camera                                            | sharp                                             |
|----|------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| S1 | 0    | <div><div>1/2 × 0.405</div><div>0.203</div></div> | 0                                                 | 0                                                 |
| S2 | 0    | 0                                                 | <div><div>1/3 × 0.405</div><div>0.135</div></div> | <div><div>1/3 × 1.099</div><div>0.366</div></div> |
| S3 | 0    | <div><div>1/3 × 0.405</div><div>0.135</div></div> | <div><div>1/3 × 0.405</div><div>0.135</div></div> | 0                                                 |

# Bag of Words vs TF-IDF

## DATASET · CLEANED

|    | text              | words |
|----|-------------------|-------|
| S1 | phone fast        | 2     |
| S2 | camera fast sharp | 3     |
| S3 | phone camera fast | 3     |

## VOCABULARY · IN HOW MANY SENTENCES

|        |        |        |        |
|--------|--------|--------|--------|
| fast   | phone  | camera | sharp  |
| 3 of 3 | 2 of 3 | 2 of 3 | 1 of 3 |

BOW · [FAST, PHONE, CAMERA, SHARP]

S1 = [1, 1, 0, 0]

S2 = [1, 0, 1, 1]

S3 = [1, 1, 1, 0]

## TF-IDF

S1 = [0, 0.203, 0, 0]

S2 = [0, 0, 0.135, 0.366]

S3 = [0, 0.135, 0.135, 0]

PART 2

# Advantages and Disadvantages of TF-IDF

ADVANTAGES

# Word Importance Is Captured

01 Intuitive

02 Fixed size input = vocabulary size

03 Word importance is captured

|    | fast | phone | camera | sharp |
|----|------|-------|--------|-------|
| S1 | 0    | 0.203 | 0      | 0     |
| S2 | 0    | 0     | 0.135  | 0.366 |
| S3 | 0    | 0.135 | 0.135  | 0     |

**fast** is in every sentence → **0**, ignored.

**sharp** is only in S2 → **highest** weight.

# How the Model Uses These Values

The model learns an **importance** for each word during training.

To predict, it does: **importance × value** for every word, then adds them up.

Anything × **0** = **0** . A word with value 0 adds nothing, so the model **ignores** it.

| S2 · CAMERA FAST SHARP · TF-IDF |                    |   |       |       |
|---------------------------------|--------------------|---|-------|-------|
| WORD                            | IMPORTANCE × VALUE |   |       | ADDS  |
| fast                            | 2                  | × | 0     | 0     |
| phone                           | 1                  | × | 0     | 0     |
| camera                          | 1                  | × | 0.135 | 0.135 |
| sharp                           | 3                  | × | 0.366 | 1.098 |
| Total score                     |                    |   |       | 1.233 |

Importances are example numbers. The model learns real ones.

# Why Ignoring a Common Word Is Good

THINK OF A CLASSROOM

Every student wears the **same uniform** . Can you find one student by their uniform?

**No. Everyone has it, so it does not help.**



SAME WITH "FAST"

"fast" is in **every** review, so it cannot tell reviews apart.

TF-IDF gives it **0** , so the model focuses on words like **sharp** that make a review different.

In bag of words, "fast" gets 1 in every sentence and adds the same amount to every score. It is just noise.

DISADVANTAGES

# Sparsity and OOV Remain

DATASET · CLEANED

|      | text               | words |
|------|--------------------|-------|
| S1   | phone fast         | 2     |
| S2   | camera fast sharp  | 3     |
| S3   | phone camera fast  | 3     |
| Test | phone battery fast |       |

VOCABULARY · IN HOW MANY SENTENCES

fast

3 of 3

phone

2 of 3

camera

2 of 3

sharp

1 of 3

- 01 Sparsity still exists

7 of our 12 values are 0. With a 50,000-word vocabulary, almost everything is 0.
- 02 Out of vocabulary (OOV)

phone ~~battery~~ fast

"battery" is not in the training vocabulary, so it is ignored.

## SUMMARY

# What to Remember

**TF** = repeats ÷ words in sentence

**IDF** =  $\ln(\text{sentences} \div \text{sentences with word})$

**TF-IDF** = **TF** × **IDF**

## ADVANTAGES

Intuitive · fixed size · word importance captured

## DISADVANTAGES

Sparsity · out of vocabulary

Next: TF-IDF practical, then Word2Vec